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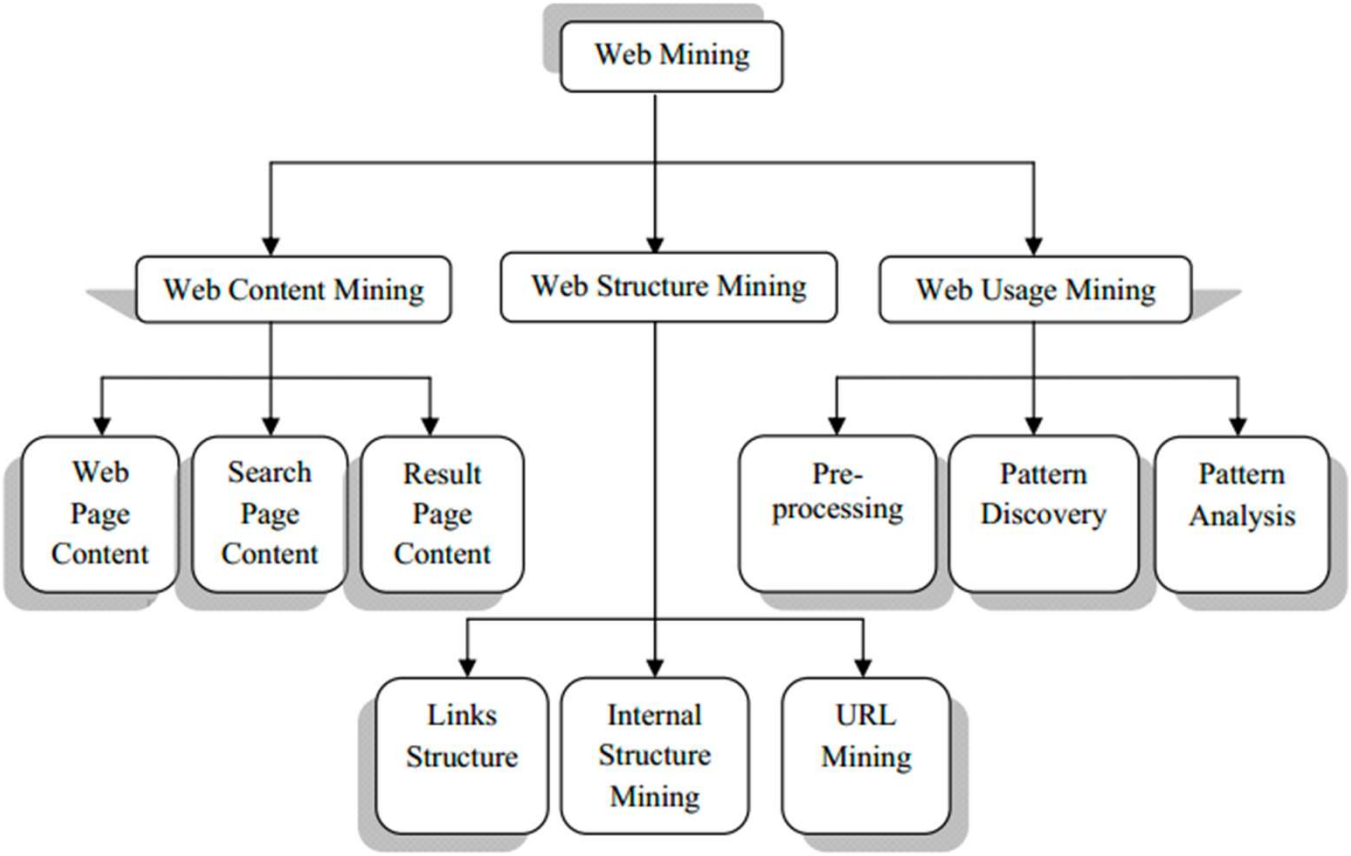
INT 7031 - Web Mining

Lecture 3: Web Data Collection & Preprocessing

Hoang-Quynh Le (PhD)

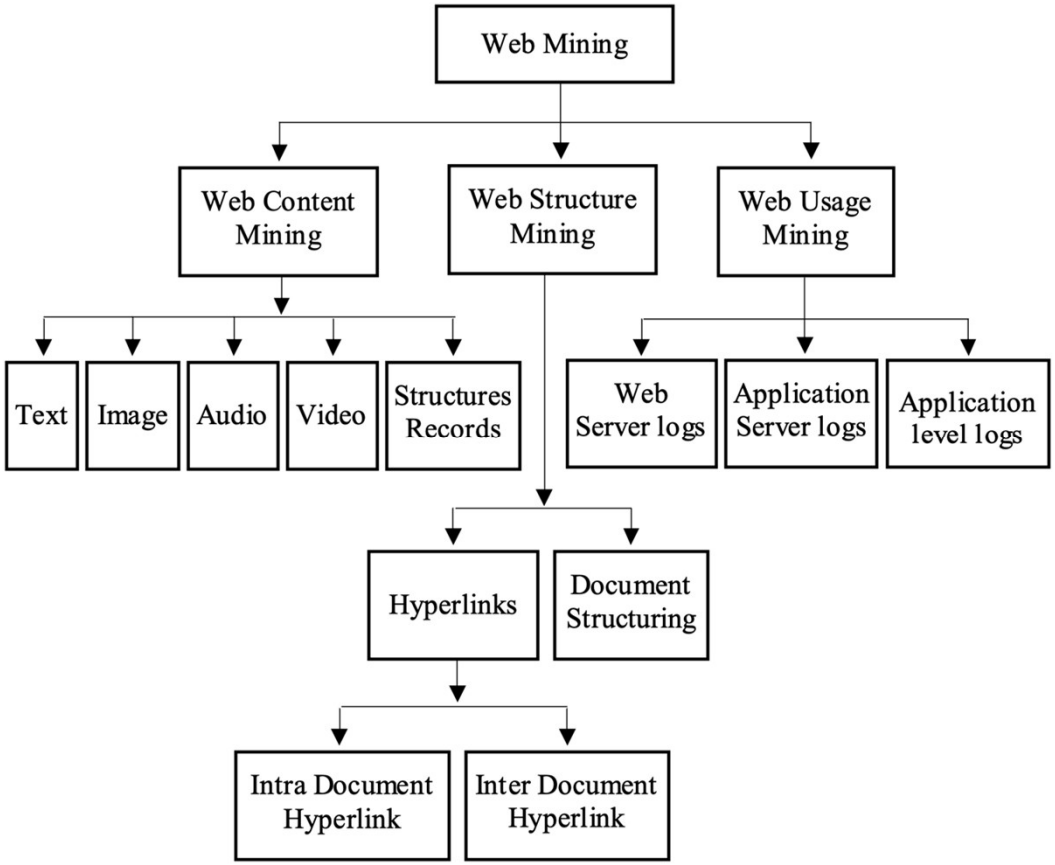
Hanoi, 03/2026

Recap: Taxonomy of Web Mining



Source: https://www.researchgate.net/figure/Taxonomy-of-Web-Mining-Source-Kavita-et-al-2011_fig1_282357293

Recap: Web Data

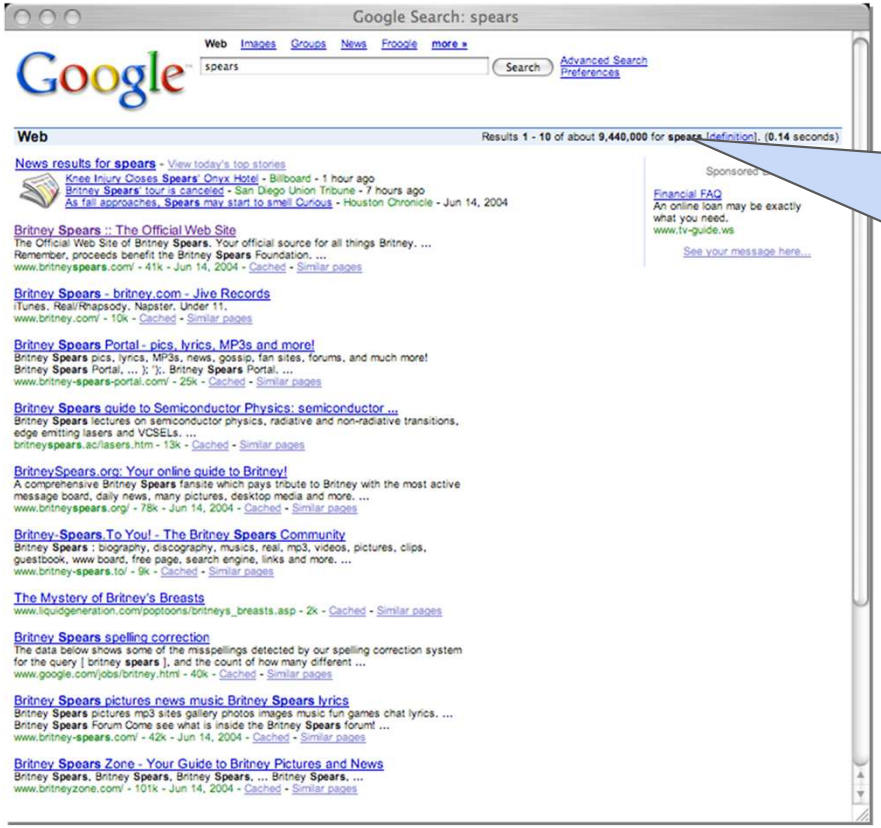


Source: Patel, Ketul B., Jignesh A. Chauhan and Jigar D. Patel. "Web Mining in E-Commerce : Pattern Discovery , Issues and Applications." (2011).

Outline

- Web Data Collection
  - Basic Crawlers
  - Large-scale Crawlers
  - Ethics & Conflicts
- Data Preprocessing
  - Data Cleaning
  - Data Integration
  - Data Transformation
  - Data Reduction

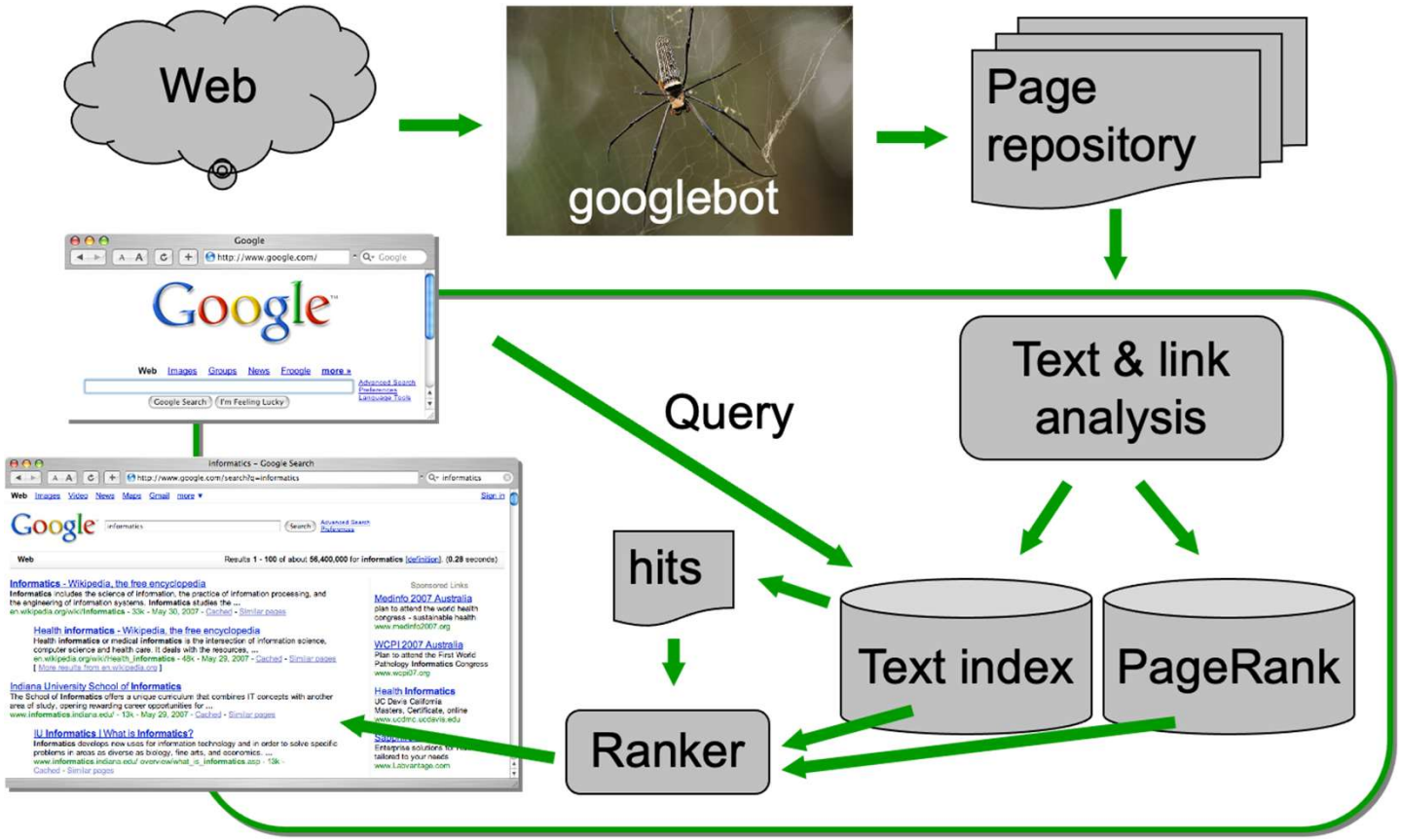
Motivation Example: Google



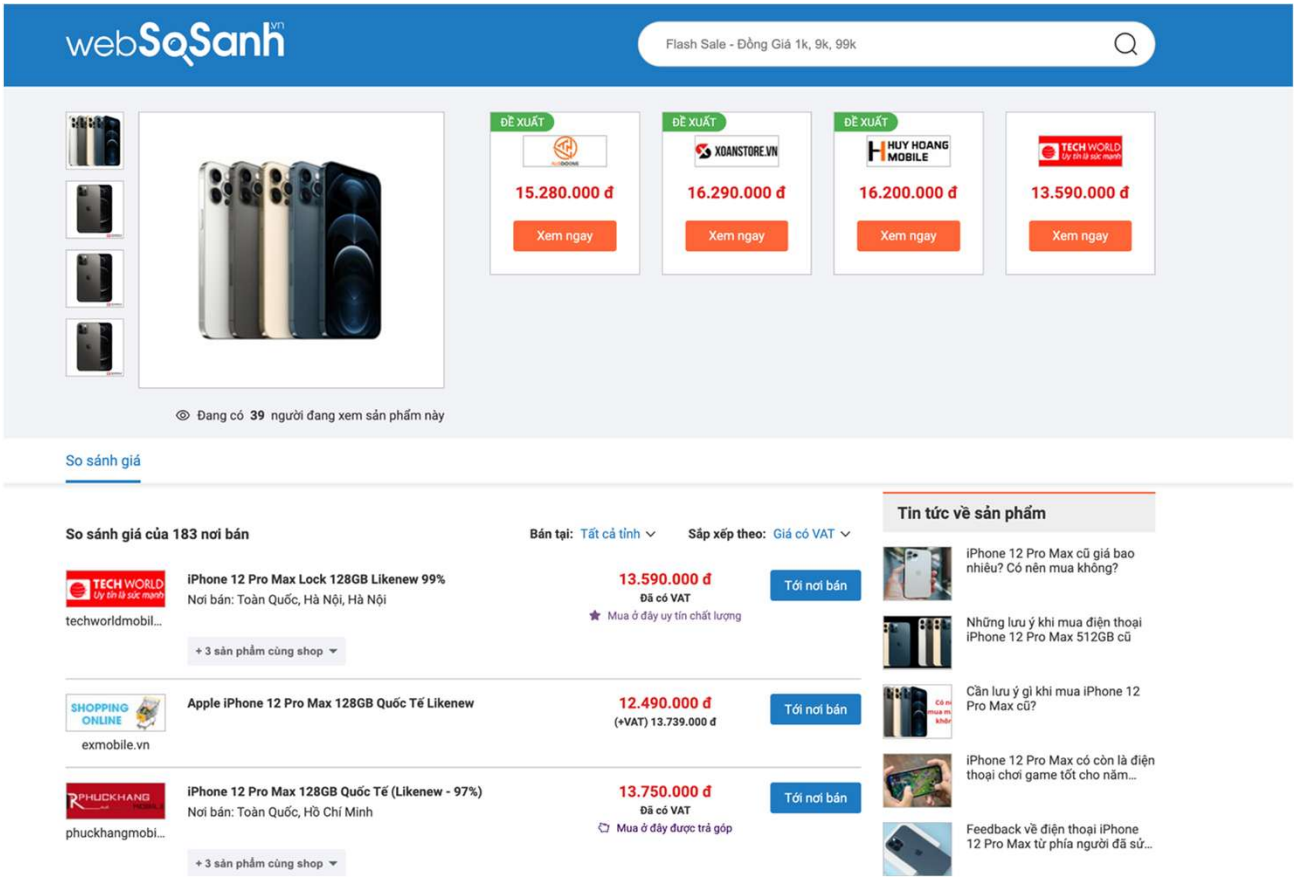
Q: How does a search engine know that all these pages contain the query terms?

A: Because all of those pages have been crawled

A Crawler in Google Search Engine

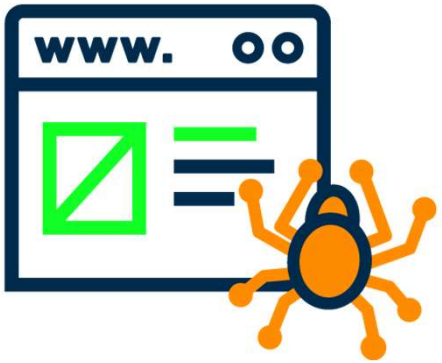


Another Example: Websosanh



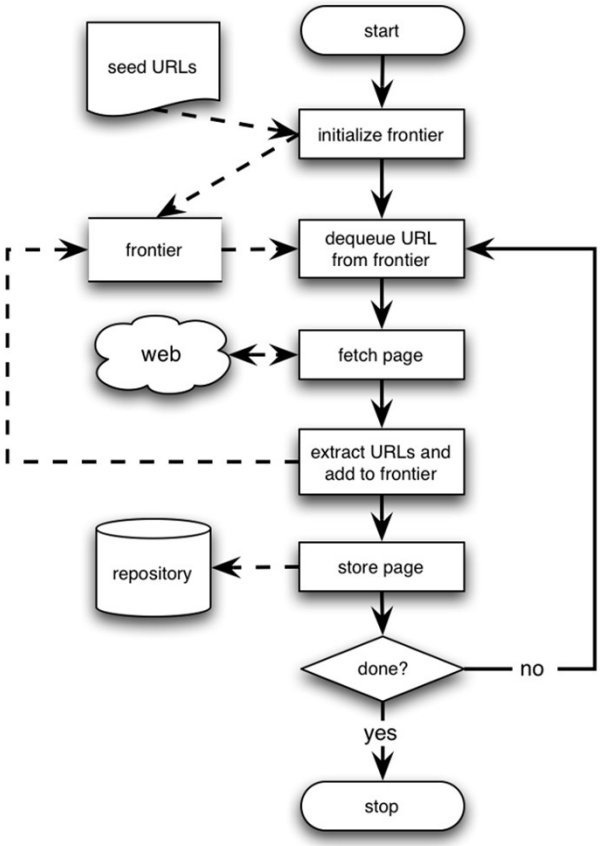
Why need Crawler(s)?

- Support universal search engines (Google, Yahoo, MSN/Windows Live, Ask, etc.)
- Vertical (specialized) search engines, e.g. news, shopping, papers, recipes, reviews, etc.
- Business intelligence: keep track of potential competitors, partners
- Monitor Web sites of interest
- …
- Web Mining Course @@

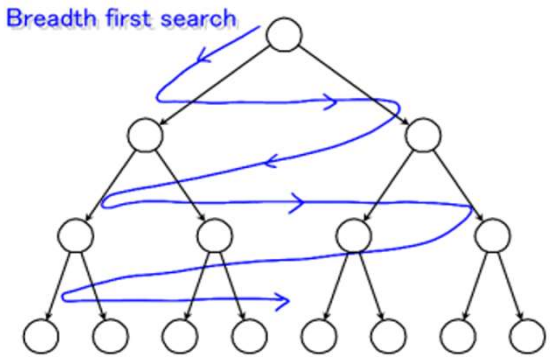


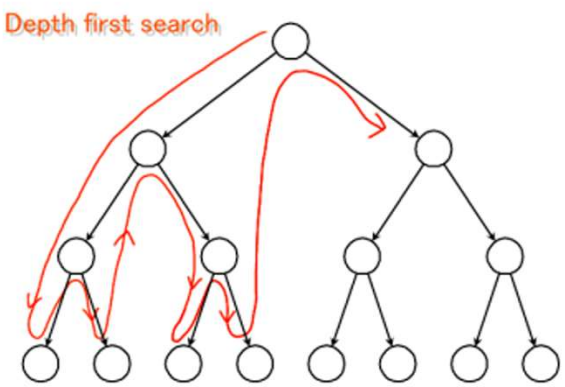
Basic Crawlers

- This is a sequential crawler
- Seeds can be any list of starting URLs
- Order of page visits is determined by frontier data structure
- Stop criterion can be anything



Graph Traversal (BFS or DFS)?

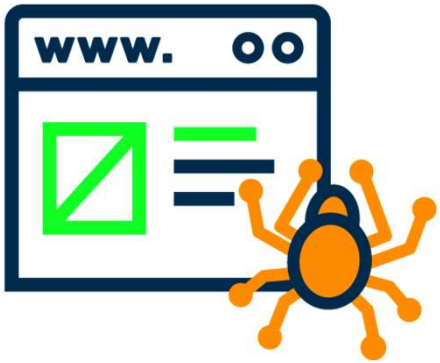




- **Breadth First Search**
  - Implemented with QUEUE (FIFO)
  - Finds pages along shortest paths
  - If we start with "good" pages, this keeps us close; maybe other good stuff…
- **Depth First Search**
  - Implemented with STACK (LIFO)
  - Wander away ("lost in cyberspace")

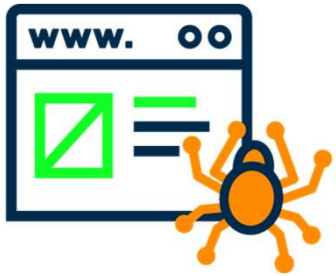
Implementation Issues

- **Don't want to fetch same page twice!**
  - Keep lookup table (hash) of visited pages
  - What if not visited but in frontier already?
- **The frontier grows very fast!**
  - May need to prioritize for large crawls
- **Fetcher must be robust!**
  - Don't crash if download fails
  - Timeout mechanism
- **Determine file type to skip unwanted files**
  - Can try using extensions, but not reliable



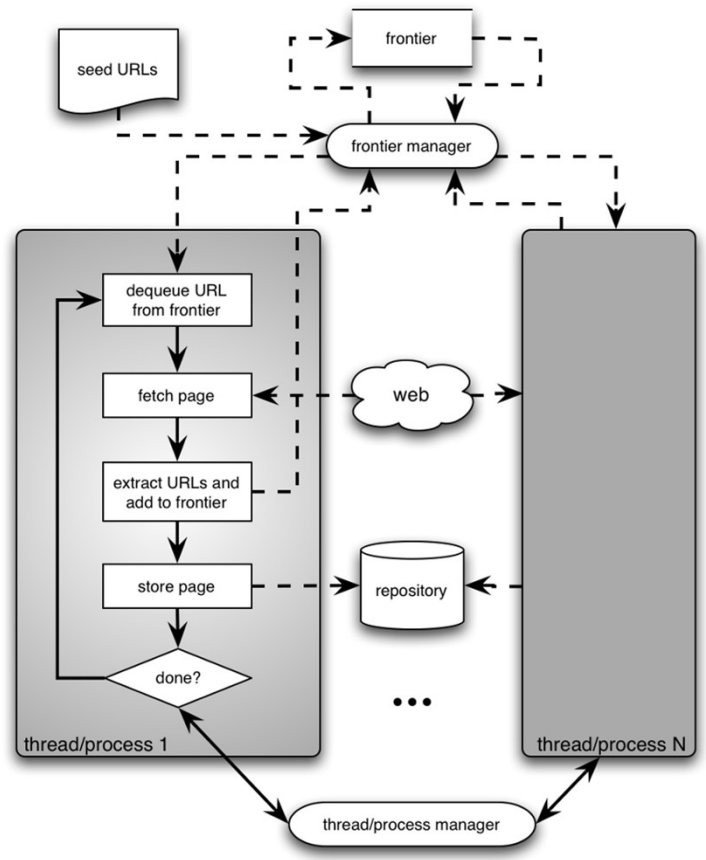
More Implementation Issues ...

- **Fetching**
  - Take care to detect and break redirection loops
  - Soft fail for timeout, server not responding, file not found, etc.
- **Static vs. dynamic pages**
  - In dynamic web pages, content of pages are different for different visitors.
- **Relative vs. Absolute URLs**
  - Base: http://www.cnn.com/linkto/
  - Relative URL: intl.html
  - Absolute URL: http://www.cnn.com/linkto/intl.html

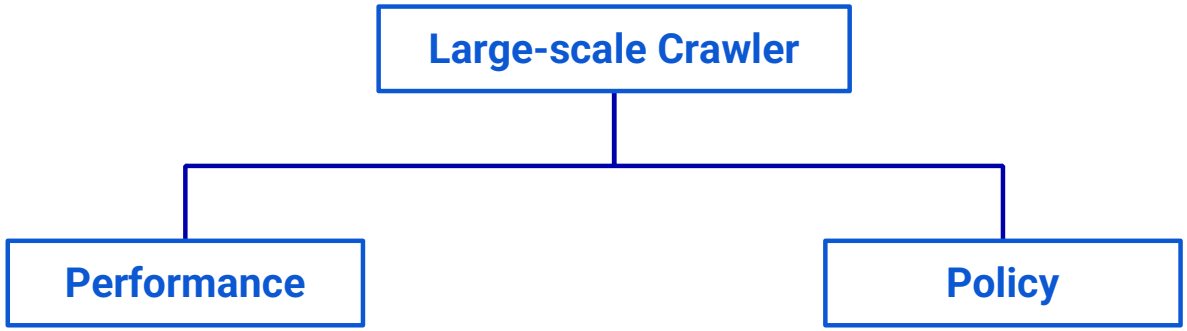


Concurrency

- **A crawler incurs several delays:**
  - Resolving the host name in the URL to an IP address using DNS
  - Connecting a socket to the server and sending the request
  - Receiving the requested page in response



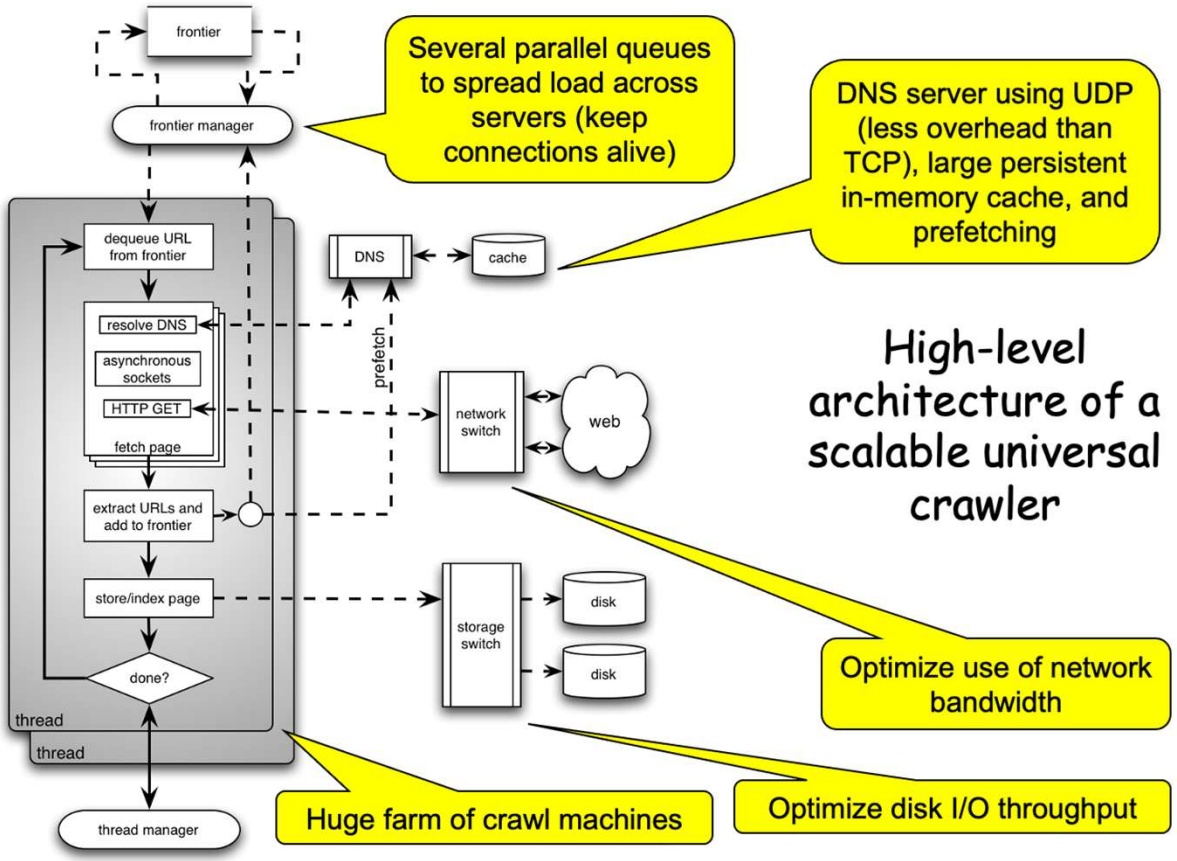
Large-Scale Crawlers



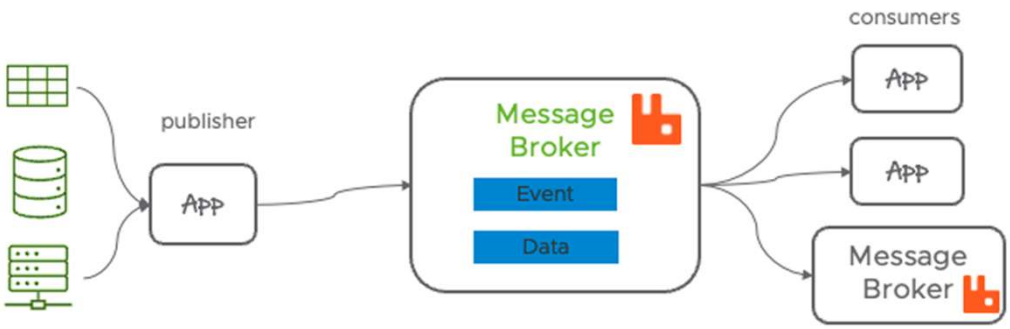
Need to scale up to billions of pages

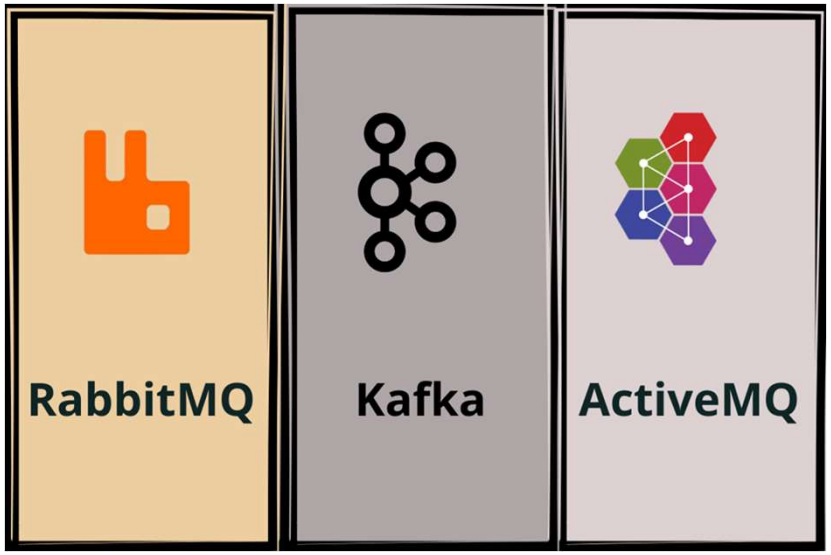
Need to trade-off coverage, freshness, and bias (e.g. toward "important" pages)

Large-Scale Crawlers



Messaging Brokers





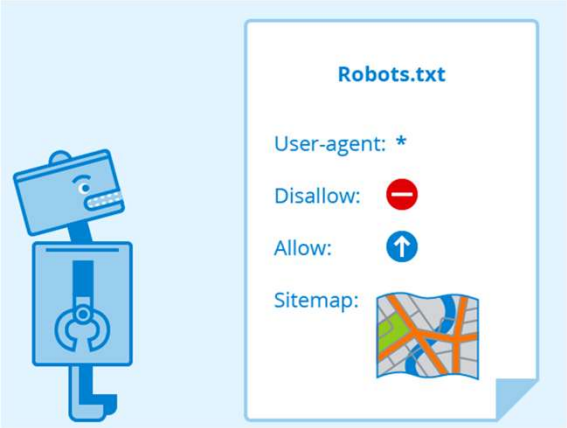
Large-Scale Crawlers - Policy

- **Coverage**
  - New pages get added all the time
  - Can the crawler find every page?
- **Freshness**
  - Pages change over time, get removed, etc.
  - How frequently can a crawler revisit ?
- **Trade-off!**
  - Focus on most “important” pages (crawler bias)?
  - “Importance” is subjective

Crawler Ethics & Conflicts

- Crawlers can cause **trouble**, even **unwillingly**, if not **properly designed** to be "**polite**" and "**ethical**"

  E.g., sending too many requests in rapid succession to a single server can amount to a Denial of Service (DoS) attack!
  - Crawler developer/admin IP address may be blacklisted
- **Solutions**:
  - Bot Identification
  - Pay attention
  - Honor the Robot Exclusion Protocol



Ref: https://www.apple.com/robots.txt

Crawler Tools

Scrapy

Apache crawler

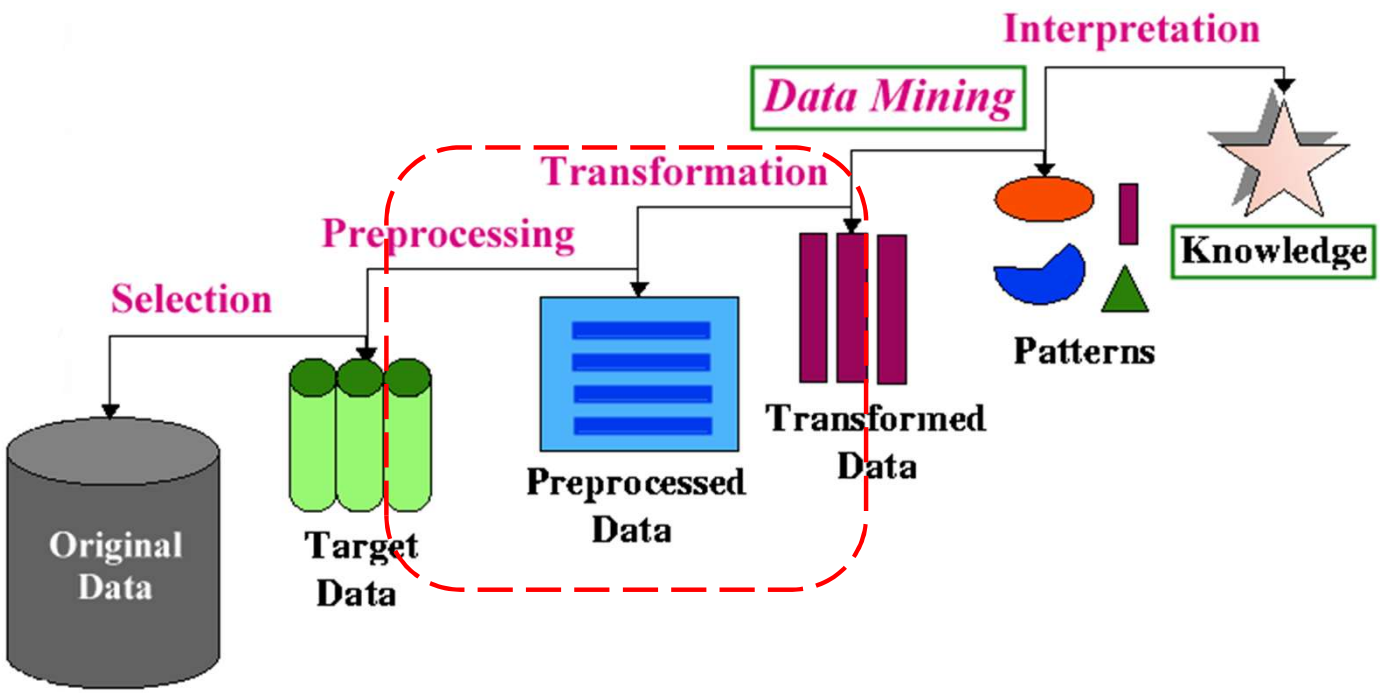
鋸
Nokogiri

Selenium 4

crawlee

View more on: https://github.com/BruceDone/awesome-crawler

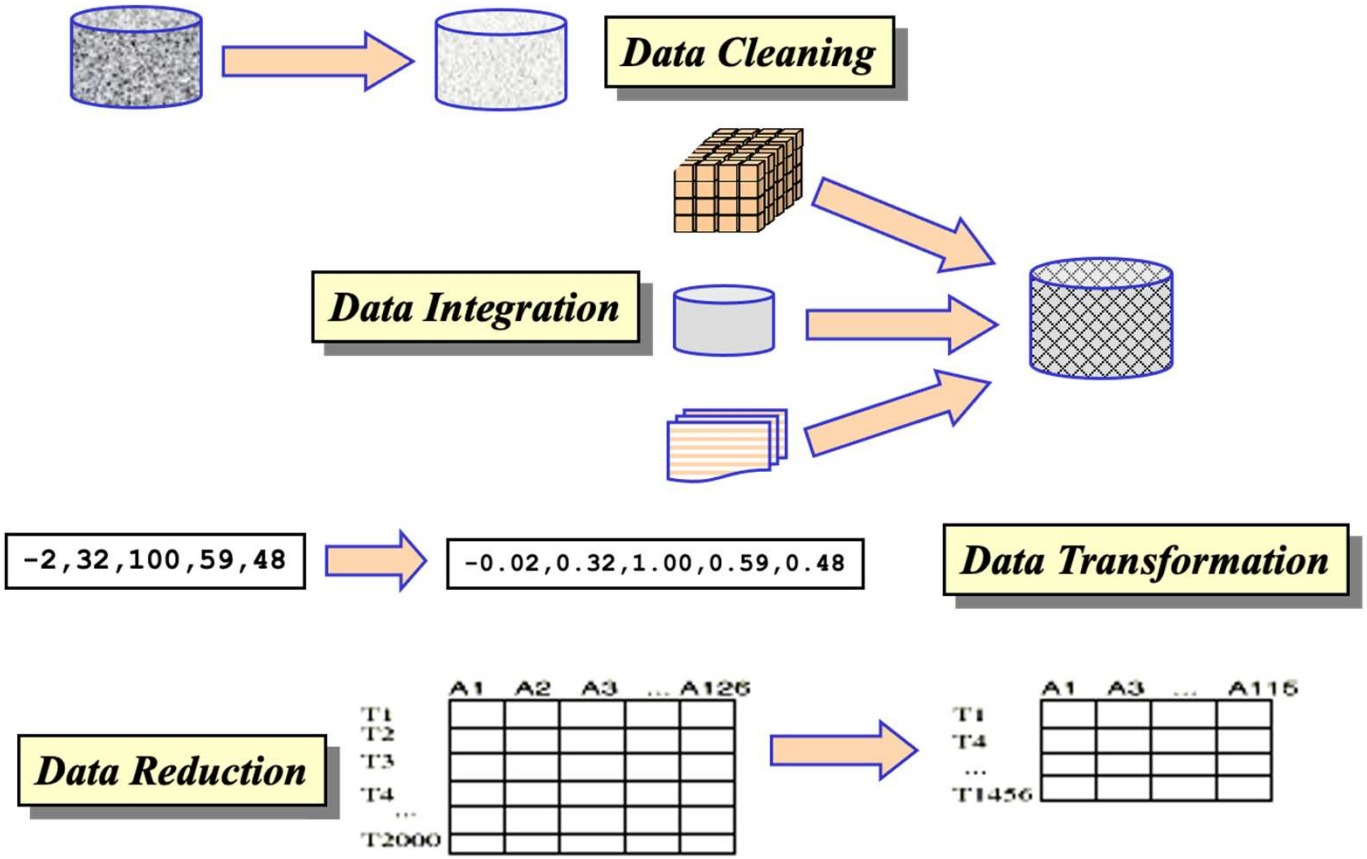
Knowledge Discovery Process



Data Preprocessing

- **Why do we need to prepare the data?** In real world applications data can be **inconsistent**, **incomplete** and/or noisy
  - Data entry, data transmission, or data collection problems
  - Discrepancy in naming conventions
  - Duplicated records
  - Incomplete or missing data
  - Contradictions in data
- **What happens when the data can not be trusted?**
  - The decision can be trusted?

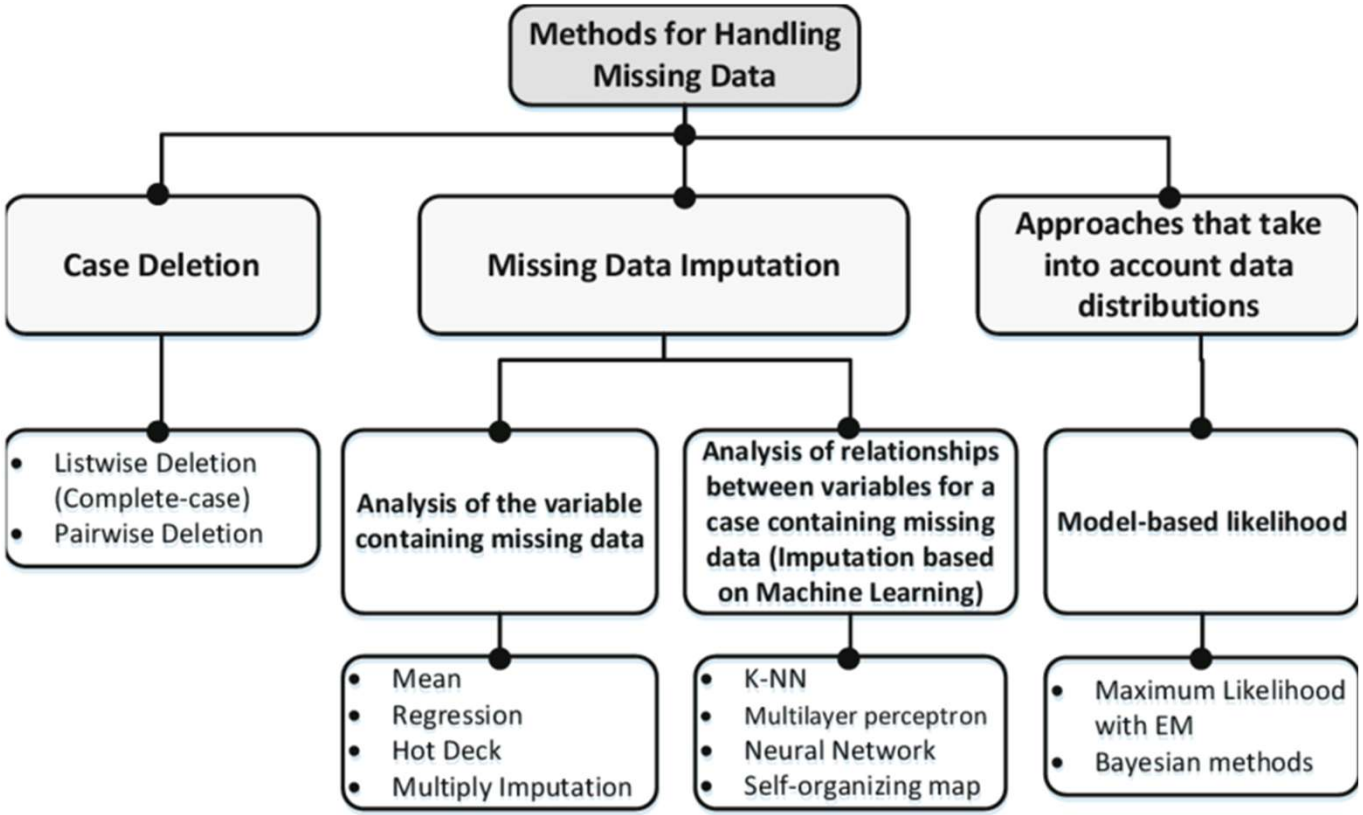
Data Preprocessing



Data Cleaning

- Real-world application data can be **incomplete**, **noisy**, and **inconsistent**
  - No recorded values for some attributes
  - Not considered at time of entry
  - Random errors
  - Irrelevant records or fields
- **Data cleaning attempts to:**
  - Fill in missing values
  - Smooth out noisy data
  - Correct inconsistencies
  - Remove irrelevant data

Dealing with Missing Values



Smoothing Noisy Data

The purpose of data smoothing is to eliminate noise and “smooth out” the data fluctuations.

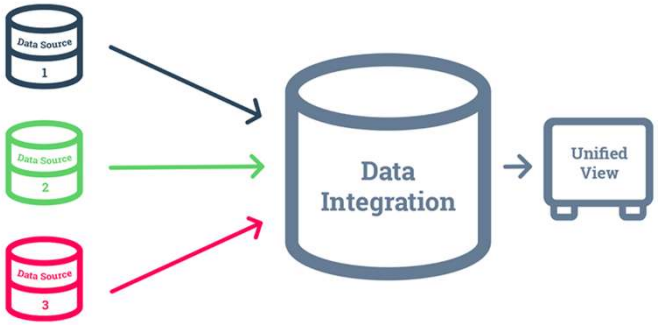
ID	Outlook	Temperature	Humidity	Windy
1	sunny	85	85	FALSE
2	sunny	80	90	TRUE
3	overcast	83	78	FALSE
4	rain	70	96	FALSE
5	rain	68	80	FALSE
6	rain	65	70	TRUE
7	overcast	58	65	TRUE
8	sunny	72	95	FALSE
9	sunny	69	70	FALSE
10	rain	71	80	FALSE
11	sunny	75	70	TRUE
12	overcast	73	90	TRUE
13	overcast	81	75	FALSE
14	rain	75	80	TRUE

ID	Temperature	
7	58	Bin1
6	65	
5	68	
9	69	Bin2
4	70	
10	71	
8	72	Bin3
12	73	
11	75	
14	75	Bin4
2	80	
13	81	
3	83	Bin5
1	85	

ID	Temperature	
7	64	Bin1
6	64	
5	64	
9	70	Bin2
4	70	
10	70	
8	73	Bin3
12	73	
11	73	
14	79	Bin4
2	79	
13	79	
3	84	Bin5
1	84	

Data Integration

- Data analysis may require a **combination** of data from **multiple sources** into a coherent data store
- **Challenges in Data Integration:**
  - Schema integration: CID = C_number = Cust-id = cust#
  - Semantic heterogeneity
  - Data value conflicts (different representations or scales, etc.)
  - Synchronization (especially important in Web usage mining)
  - Redundant attributes



Data Transformation: Normalization

- **Min-max normalization:**

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

- **z-score normalization:**

$$z = \frac{\overset{\text{Score}}{x} - \overset{\text{Mean}}{\mu}}{\underset{\text{SD}}{\sigma}}$$

- **Normalization by decimal scaling**

$$v' = \frac{v}{10^j}$$

ID	Gender	Age	Salary
1	F	27	19,000
2	M	51	64,000
3	M	52	100,000
4	F	33	55,000
5	M	45	45,000

ID	Gender	Age	Salary
1	1	0.00	0.00
2	0	0.96	0.56
3	0	1.00	1.00
4	1	0.24	0.44
5	0	0.72	0.32

Data Transformation: Discretization

- **03 Types of attributes**
  - nominal - values from an unordered set (also “categorical” attributes)
  - ordinal - values from an ordered set
  - numeric/continuous - real numbers (but sometimes also integer values)
- **Discretization is used to reduce the number of values for a given continuous attribute.**
  - usually done by dividing the range of the attribute into intervals
  - interval labels are then used to replace actual data values
- **Discretization can also be used to generate concept hierarchies**
  - reduce the data by collecting and replacing low level concepts (e.g., numeric values for “age”) by higher level concepts (e.g., “young”, “middle aged”, “old”)

Data Transformation: Discretization - Example

Humidity
85
90
78
96
80
70
65
95
70
80
70
90
75
80

Low = 60-69
Normal = 70-79
High = 80+

Humidity
High
High
Normal
High
High
Normal
Low
High
Normal
High
Normal
High
Normal
High

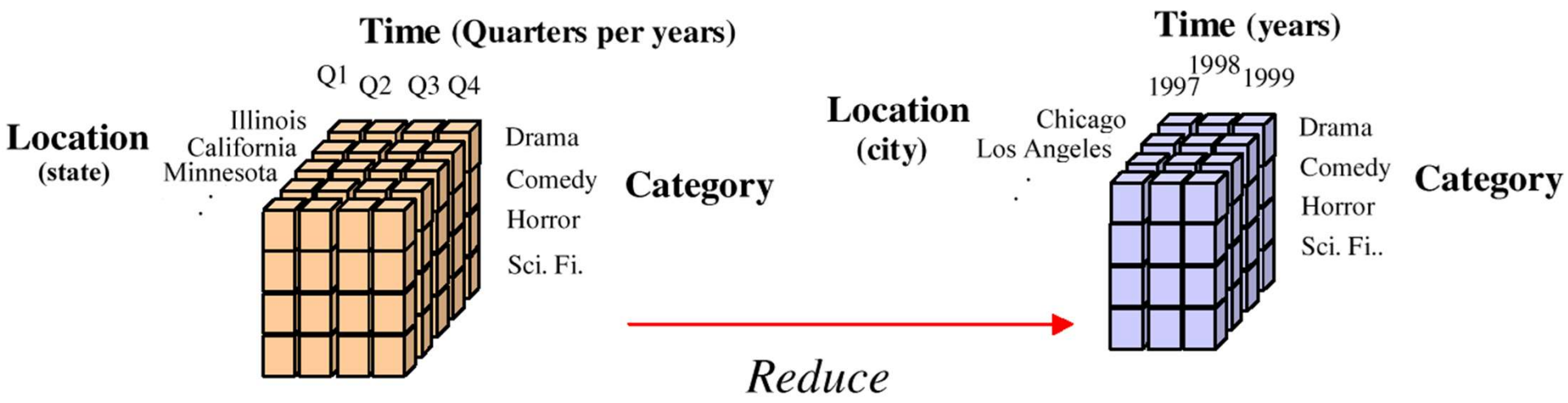
Data Transformation: Discretization - Methods

- **Binning**
  - Top-down split, unsupervised
- **Histogram analysis**
  - Top-down split, unsupervised
- **Clustering analysis**
  - Unsupervised, top-down split or bottom-up merge
- **Decision-tree analysis**
  - Supervised, top-down split
- **Correlation (e.g., x2) analysis**
  - Unsupervised, bottom-up merge

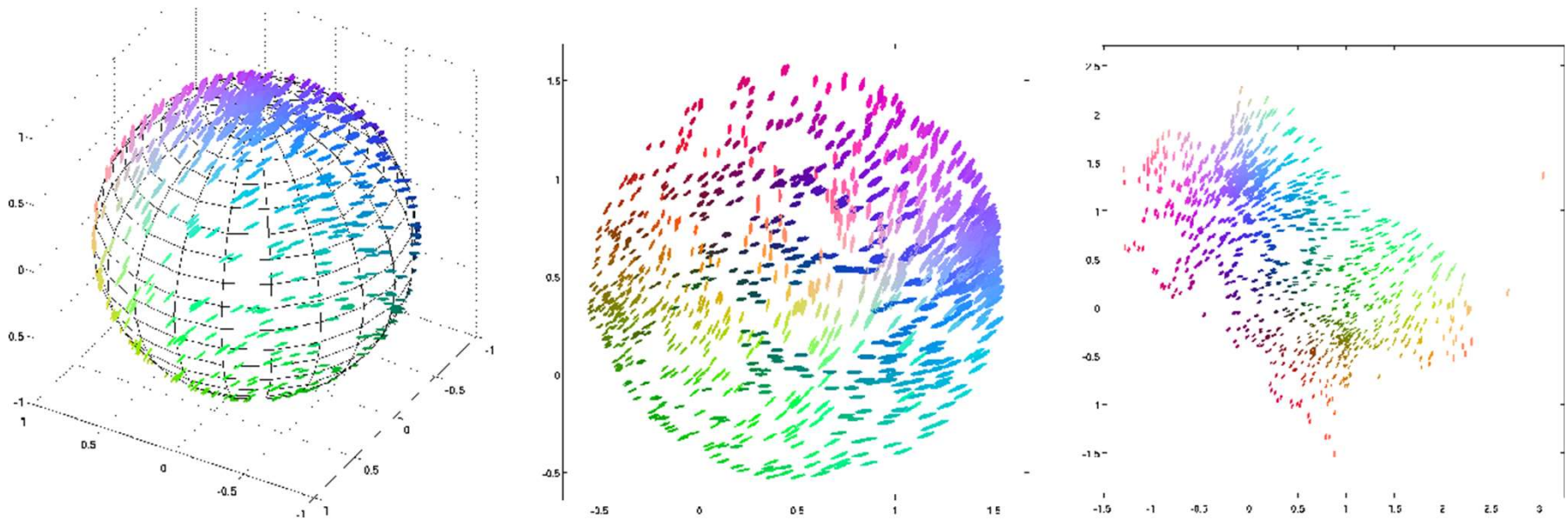
Data Reduction

- Data is often **too large**; **reducing** data can **improve performance**
- Data reduction consists of reducing the **representation** of the data set while **producing** the **same** (or almost the same) **results**
- **Data reduction includes:**
  - Data cube aggregation
  - Dimensionality reduction
  - Discretization
  - Numerosity reduction
    - Regression
    - Histograms
    - Clustering
    - Sampling

Data Cube Aggregation - Example



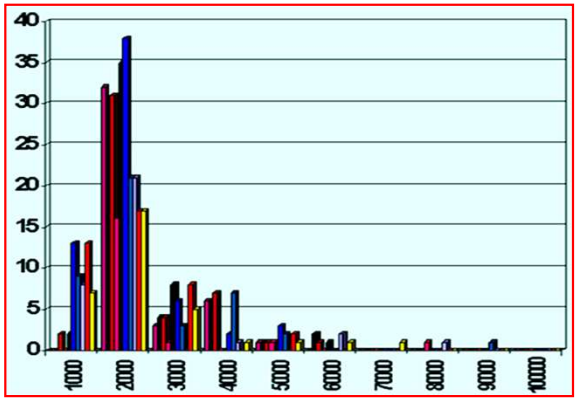
Dimensionality Reduction - Example

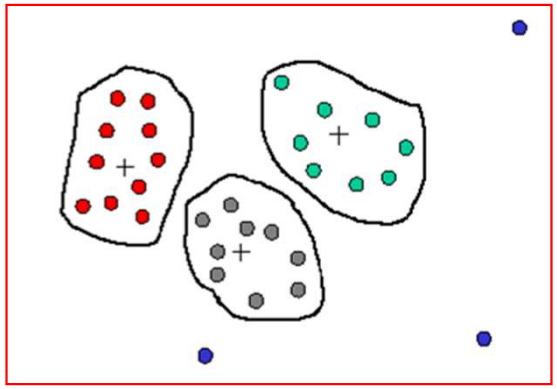


Popular Techniques: PCA, SVD, MF

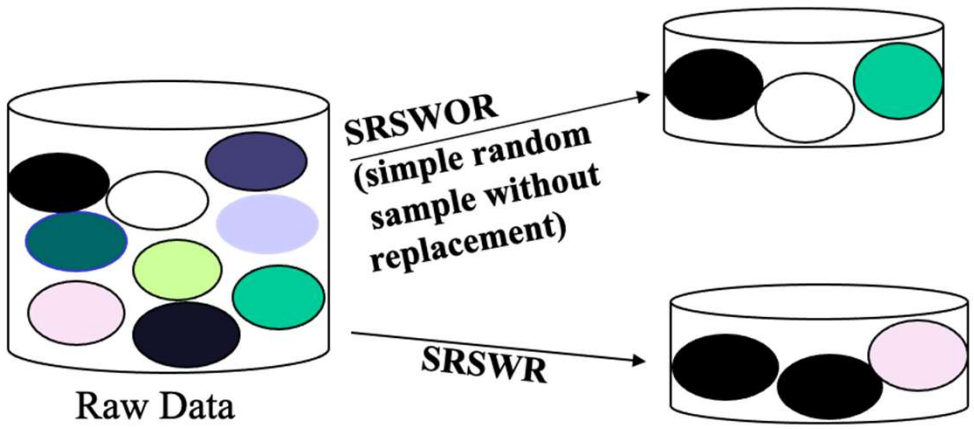
Numerocity Reduction

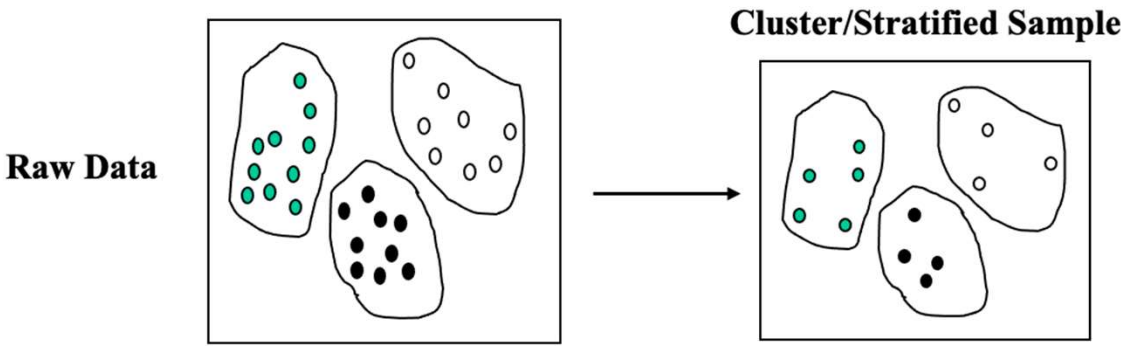
- **Reduction via histograms:**
  - Divide data into buckets and store representation of buckets (sum, count, etc.)
- **Reduction via clustering**
  - Partition data into clusters based on “closeness” in space
  - Retain representatives of clusters (centroids) and outliers
- **Reduction via sampling**
  - Random sampling can produce poor results
  - Stratified sample (stratum = group based on attribute value)





Sampling Techniques





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  - Data Cleaning
  - Data Integration
  - Data Transformation
  - Data Reduction

Thank you

Email me
lhquynh@vnu.edu.vn